CHAPTER 12

ORDINAL NUMBERS

The Axiom of Replacement finds its most important applications in von Neumann's beautiful theory of ORDINAL NUMBERS, and in the construction of the CUMULATIVE HIERARCHY of pure, grounded sets. One can live without knowing the ordinals, to be sure, but not as well: they bring many gifts, among them true cardinal numbers which give substance to the "virtual" theory of equinumerosities with which we have been making do. The Cumulative Hierarchy extends the iteration of the power operation we have used to construct V_ω "as far as it will go" and presents the pure, grounded sets as the most compelling intuitive understanding of what sets really are. It is not so clear one can live without knowing that, not among set theorists, at any rate.

Cantor describes his conception of "ordinal types" just a few pages after the definition of cardinals quoted in **4.19**, and in a very similar vein.

> Every ordered set U has a definite 'ordinal type', ... which we will denote by $\overline{U}$. By this we understand the general concept which results from U if we only abstract from the nature of the elements u, and retain the order or precedence among them. Thus the ordinal type $\overline{U}$ is itself an ordered set whose elements are units which have the same order of precedence amongst one another as the corresponding elements of U, from which they are derived by abstraction. ... A simple consideration shows that two ordered sets have the same ordinal type if, and only if, they are similar, so that of the two formulas $U =_o V$, $\overline{U} = \overline{V}$, one is always a consequence of the other.

Cantor is speaking about arbitrary linearly ordered sets, but we will consider here only the problem of defining "ordinal types" for well ordered sets. He states explicitly the first key property

$$U =_o \overline{U} \tag{12-1}$$

of the ordinal assignment operation, and argues for

$$U =_o V \Longrightarrow \overline{U} = \overline{V}. \tag{12-2}$$

Cantor's implied "simple consideration" for (12-2) should also justify (for well ordered sets) the stronger implication

$$U \leq_o V \Longrightarrow \overline{U} \sqsubseteq \overline{V}; \tag{12-3}$$

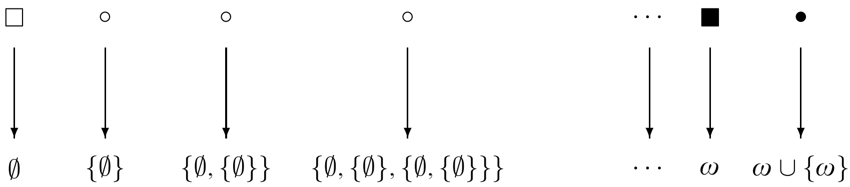

FIGURE **12.1**. The von Neumann map of $U \ : \ 0_U, 1_U, 2_U, \dots, \omega_U, S_U(\omega_U)$.

because the position of a point x in a well ordered set depends only on the points preceding it, so the "unit" $\overline{x}$ abstracted from x and coding its place in U should depend only on the initial segment $\mathbf{seg}_U(x)$. Thus, the problem of representing Cantor's conception of ordinals in axiomatic set theory comes down to the following: *can we assign a well ordered set $\overline{U}$ to each well ordered set U, so that* (12-1) *and* (12-3) *hold?* Von Neumann's ingenious idea is to define $\overline{U}$ by replacing *recursively* each member of U by the set of its predecessors. The construction is captured exactly by the Mostowski Collapsing Lemma **11.36**.

12.1. Ordinal numbers. The **von Neumann map** of a well ordered set U is the unique decoration of the associated grounded graph $(\mathrm{Field}(U), >_U)$, so that by (11-29),

$$\boldsymbol{v}_U(x) = \{\boldsymbol{v}_U(y) \mid y <_U x\}, \quad (x \in \mathrm{Field}(U)). \tag{12-4}$$

We define the **ordinal number** of U to be the image

$$\mathrm{ord}(U) =_{\mathrm{df}} \boldsymbol{v}_U[\mathrm{Field}(U)] \tag{12-5}$$

of its von Neumann map, and we set

$$\mathrm{ON}(\alpha) \iff \alpha \in \mathrm{ON} \iff_{\mathrm{df}} (\exists \text{ well ordered } U)[\alpha = \mathrm{ord}(U)]. \tag{12-6}$$

Suppose for example that

$$U \ : \ 0_U, 1_U, 2_U, \dots, \omega_U, S_U(\omega_U)$$

is a well ordered set with least element 0_U, next 1_U, ..., first limit point ω_U, followed by the last (largest) point $S_U(\omega_U)$. We compute the values of its von Neumann surjection by repeated applications of (12-4) (skipping the subscript):

$$\begin{array}{rcll}
\boldsymbol{v}(0_U) & = & \{\boldsymbol{v}(x) \mid x < 0_U\} & = \emptyset & = 0, \\
\boldsymbol{v}(1_U) & = & \{\boldsymbol{v}(x) \mid x < 1_U\} & = \{\emptyset\} & = 1, \\
\boldsymbol{v}(2_U) & = & \{\boldsymbol{v}(x) \mid x < 2_U\} & = \{\emptyset, \{\emptyset\}\} & = 2, \\
\boldsymbol{v}(3_U) & = & \{\boldsymbol{v}(x) \mid x < 3_U\} & = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\} & = 3, \\
 & \vdots & & & \\
\boldsymbol{v}(\omega_U) & = & \{\boldsymbol{v}(x) \mid x < \omega_U\} & = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}, \dots\} & = \omega \\
\boldsymbol{v}(S_U(\omega_U)) & = & \{\boldsymbol{v}(x) \mid x < S_U(\omega_U)\} & = \omega \cup \{\omega\} &
\end{array}$$

$$\boldsymbol{v}[U] = \{0, 1, 2, \dots, \omega, \omega \cup \{\omega\}\}.$$

These computations are special cases of the following, general facts:

$$\begin{tikzcd}
U \arrow[rr, "\pi"] \arrow[d, "\boldsymbol{v}_U"'] & & V \arrow[d, "\boldsymbol{v}_V"] \\
\mathrm{ord}(U) \arrow[rr, "id(x)=x"] & & \mathrm{ord}(V)
\end{tikzcd}$$

FIGURE **12.2**. The von Neumann map under initial similarities.

12.2. Exercise. *If 0_U is the least point in a well ordered set U, then $\boldsymbol{v}_U(0_U) = \emptyset$; and if $S(x)$ is the successor of x in U, then*

$$\boldsymbol{v}_U(S(x)) = \boldsymbol{v}_U(x) \cup \{\boldsymbol{v}_U(x)\}.$$

12.3. Exercise. *If x is a limit point in a well ordered set U, then $\emptyset \in \boldsymbol{v}_U(x)$ and*

$$\alpha \in \boldsymbol{v}_U(x) \Longrightarrow \alpha \cup \{\alpha\} \in \boldsymbol{v}_U(x).$$

12.4. Exercise. *If ω_U is the first limit point in a well ordered set U, then*

$$\omega = \boldsymbol{v}_U(\omega_U) = \bigcap \{X \mid \emptyset \in X \;\&\; (\forall \alpha \in X)[\alpha \cup \{\alpha\} \in X]\}, \tag{12-7}$$

and, in particular, $\omega = \boldsymbol{v}_U(\omega_U)$ is independent of the particular well ordered set U used to compute it.

This exercise makes clear about ω_U what is evident about $0_U, 1_U, \ldots$ in Figure **12.1**; the value $\boldsymbol{v}_U(x)$ is independent of the particular element $x \in U$, and depends only on the place of x in U, whether it is the first element, the fifth, the first limit point or whatever. This is a general fact about the von Neumann map, which we can make precise as follows.

12.5. Lemma (First Ordinal Property). *If $\pi : U \rightarrowtail\!\!\!\!\twoheadrightarrow \pi[U] \sqsubseteq V$ is an initial similarity from U into another well ordered set V, then the diagram in Figure* **12.2** *commutes, i.e.,*

$$\boldsymbol{v}_V(\pi(x)) = \boldsymbol{v}_U(x) \quad (x \in U). \tag{12-8}$$

PROOF. Towards a contradiction, let x be the least element of U such that $\boldsymbol{v}_V(\pi(x)) \neq \boldsymbol{v}_U(x)$, and compute:

$$\begin{aligned}
\boldsymbol{v}_V(\pi(x)) &= \{\boldsymbol{v}_V(y) \mid y <_V \pi(x)\} && \text{by definition,} \\
&= \{\boldsymbol{v}_V(\pi(t)) \mid t <_U x\} && \text{because } \pi \text{ is initial,} \\
&= \{\boldsymbol{v}_U(t) \mid t <_U x\} && \text{by the choice of } x, \\
&= \boldsymbol{v}_U(x),
\end{aligned}$$

contradicting the choice of x. The key step here is the second one, where we used the fact that an initial similarity "has no gaps" in its image, so that each $y <_V \pi(x)$ is $\pi(t)$ for some $t <_U x$. ⊣

12.6. Exercise. *For any two well ordered sets U, V,*

$$U \leq_o V \Longrightarrow \mathrm{ord}(U) \subseteq \mathrm{ord}(V);$$

and so, if $U =_o V$, then $\mathrm{ord}(U) = \mathrm{ord}(V)$.

12.7. Lemma (Second ordinal property). *For each well ordered set U and each $x \in U$,*

$$\boldsymbol{v}_U(x) = \mathrm{ord}(\mathbf{seg}_U(x)). \tag{12-9}$$

As a consequence, each von Neumann value $\boldsymbol{v}_U(x)$ is an ordinal number, and conversely, each ordinal number α is the von Neumann value $\boldsymbol{v}_U(x)$ of some point in a well ordered set.

Put another way: each member of an ordinal is an ordinal and every ordinal is a member of an ordinal.

Proof. If we apply Lemma **12.5** with $\mathbf{seg}_U(x)$ for U, U for V and the identity initial similarity $\pi : \mathbf{seg}_U(x) \rightarrowtail U$, we get:

$$\boldsymbol{v}_{\mathbf{seg}_U(x)}(y) = \boldsymbol{v}_{\mathbf{seg}_U(x)}(\pi(y)) = \boldsymbol{v}_U(y) \qquad (y <_U x),$$

so that, in the end

$$\boldsymbol{v}_U(x) = \{\boldsymbol{v}_U(y) \mid y <_U x\} = \{\boldsymbol{v}_{\mathbf{seg}_U(x)}(y) \mid y <_U x\} = \mathrm{ord}(\mathbf{seg}_U(x)).$$

For the second claim, let $V = \mathrm{Succ}(U)$ be the next well ordered set to U, with t added on top, as in **7.16**: now $U = \mathbf{seg}_V(t)$, and so $\mathrm{ord}(U) = \boldsymbol{v}_V(t)$. ⊣

Thus, we can think of ordinal numbers as standing either for lengths of well ordered sets, or for places of points in a well ordered set. The latter agrees more with the use of ordinals in ordinary language, where "first", "second", ... customarily describe the place of objects in a sequence.

12.8. Exercise (The finite von Neumann ordinals). *If $\leq_{\mathbb{N}}$ is the usual ordering of the set $\mathbb{N}$ of natural numbers, then*

$$\mathrm{ord}(\mathbb{N}, \leq_{\mathbb{N}}) = \omega,$$

as this is defined in (12-7). *Moreover, if we set*

$$S_\omega(n) = n \cup \{n\} \quad (n \in \omega),$$

then $(\omega, \emptyset, S_\omega)$ is a Peano system, and $\boldsymbol{v}_{\mathbb{N}} : \mathbb{N} \rightarrowtail\!\!\!\!\rightarrow \omega$ is the unique (*by Theorem* **5.4**) *isomorphism of the natural numbers with ω.*

It is usual in advanced set theory to take $(\omega, \emptyset, S)$ as *the* Peano system we fixed in **5.9**, i.e., to identify $\mathbb{N}$ with ω. This is sometimes convenient, but neither necessary, nor especially natural—it is hard to argue that $\{\emptyset, \{\emptyset\}\}$ is a better representation of the number 2 than Zermelo's $\{\{\emptyset\}\}$ which comes from the proof of Theorem **5.3**, or the third member of any Peano system.

Next comes the basic fact about ordinal numbers.

12.9. Lemma (Third ordinal property). *Each ordinal number α is well ordered by the relation*

$$u \leq_\alpha v \iff_{\mathrm{df}} u = v \vee u \in v \quad (u, v \in \alpha); \tag{12-10}$$

and if $\alpha = \mathrm{ord}(U)$ for a well ordered set U, then the von Neumann map $\boldsymbol{v}_U : U \twoheadrightarrow \alpha$ is a similarity.

It follows that every well ordered set is similar with an ordinal number, and every well orderable set is equinumerous with an ordinal.

PROOF. Notice first that by Lemma **11.36**, each $\alpha = \boldsymbol{v}_U[U]$ is a transitive, pure and grounded set, and (writing $\boldsymbol{v}$ for $\boldsymbol{v}_U$),

$$x <_U y \Longrightarrow \boldsymbol{v}(x) \in \boldsymbol{v}(y). \tag{12-11}$$

Moreover,

$$x \neq y \Longrightarrow \boldsymbol{v}(x) \neq \boldsymbol{v}(y), \quad (x, y \in U);$$

because if $x \neq y$, then either $x <_U y$ or $y <_U$, so that by (12-11), either $\boldsymbol{v}(x) \in \boldsymbol{v}(y)$ or $\boldsymbol{v}(y) \in \boldsymbol{v}(x)$; and, in either case, we can't have $\boldsymbol{v}(x) = \boldsymbol{v}(y)$ since α is grounded. Thus $\boldsymbol{v} : U \rightarrowtail\!\!\!\twoheadrightarrow \alpha$ is an injection, hence a bijection, and the relation $\leq_\alpha$ is the image of $\leq_U$ by $\boldsymbol{v}$, i.e.,

$$x \leq_U y \iff \boldsymbol{v}(x) \leq_\alpha \boldsymbol{v}(y) \quad (x, y \in U);$$

so $\leq_\alpha$ well orders α, and $\boldsymbol{v}$ is a similarity.

The last claim follows because similarities are bijections. ⊣

This remarkable result says, in effect, that there exist sufficiently long $\in$-chains to mirror every wellordering, and it is a characteristic consequence of the Replacement Axiom. As we have been doing with structured sets throughout, by "the ordinal α" we will mean ambiguously the set α or the well ordered set $(\alpha, \leq_\alpha)$, so that, for example, Lemma **12.9** is expressed simply by

$$U =_o \mathrm{ord}(U).$$

12.10. Exercise. *For every ordinal number* α, $\mathrm{ord}(\alpha) = \alpha$.

12.11. Corollary (Characterization of ordinals). *A set* α *is an ordinal if and only if it is transitive, pure, grounded and* $\in$*-connected, i.e.,*

$$\mathrm{ON}(\alpha) \iff [x = y \vee x \in y \vee y \in x] \qquad (x, y \in \alpha).$$

PROOF. Every ordinal has these properties, by Lemmas **11.36** and **12.9**. For the converse, suppose A is transitive, pure, grounded and $\in$-connected, and set (as if A were an ordinal)

$$x \leq_A y \iff x = y \vee x \in y, \quad (x, y \in A).$$

This relation (easily) well orders A, so let $\boldsymbol{v}_A : A \rightarrowtail\!\!\!\twoheadrightarrow \mathrm{ord}(A, \leq_A)$ be its von Neumann map. We now claim that $\boldsymbol{v}_A$ is the identity map; if not, let x be $\leq_A$-least such that $\boldsymbol{v}_A(x) \neq x$, and compute:

$$\begin{aligned} \boldsymbol{v}_A(x) &= \{\boldsymbol{v}(y) \mid y <_A x\} \\ &= \{y \mid y <_A x\} && \text{(by the choice of } x) \\ &= \{y \mid y \in x\} && \text{(by the definition of } \leq_A) \\ &= x && \text{(because } x \text{ has no atoms)}, \end{aligned}$$

contradicting the choice of x. Thus $\mathrm{ord}(A, \leq_A) = \boldsymbol{v}_A[A] = A$, and A is an ordinal. ⊣

This characterization of ordinal numbers is especially simple when we assume that all sets are grounded and pure, as we do in **ZFC**: a *set then is an ordinal exactly when it is transitive and* $\in$*-connected.*

The three basic ordinal properties also give a strong solution to Cantor's problem of defining ordinal types for well ordered sets, as we formulated it in (12-1) – (12-3).

12.12. Theorem. *The definite operation* $U \mapsto \mathrm{ord}(U)$ *on well ordered sets satisfies the following conditions*:

$$U =_o \mathrm{ord}(U), \tag{12-12}$$

$$U \leq_o V \Longrightarrow \mathrm{ord}(U) \sqsubseteq \mathrm{ord}(V), \tag{12-13}$$

$$\mathrm{ON}(\alpha) \Longrightarrow \alpha = \{\beta \in \mathrm{ON} \mid \beta <_o \alpha\}. \tag{12-14}$$

PROOF. The first property (12-12) is a restatement of Lemma **12.9**.

To prove (12-13), suppose $\pi : U \rightarrowtail\!\!\!\!\twoheadrightarrow \pi[U] \sqsubseteq V$ is an initial similarity. By Lemma **12.7**, taking images,

$$\boldsymbol{v}_V[\pi[U]] = \boldsymbol{v}_U[U] = \mathrm{ord}(U);$$

and since $\boldsymbol{v}_V$ is a similarity of V with $\mathrm{ord}(V)$, it carries initial segments onto initial segments, so that

$$\mathrm{ord}(U) = \boldsymbol{v}_U[U] = \boldsymbol{v}_V[\pi[U]] \sqsubseteq \mathrm{ord}(V).$$

Finally, for (12-14), if $\alpha = \mathrm{ord}(U)$, then:

$$\begin{aligned} \alpha &= \{\boldsymbol{v}_U(y) \mid y \in U\} \\ &= \{\mathrm{ord}(\mathbf{seg}_U(y)) \mid y \in U\} \quad \text{(by Lemma } \mathbf{12.7}) \\ &= \{\beta \in \mathrm{ON} \mid \beta <_o \alpha\}, \end{aligned}$$

the last because the well ordered sets which are $<_o U$ are exactly those similar with the proper initial segments of U. ⊣

12.13. Exercise. *For any two well ordered sets U, V,*

$$U =_o V \iff \mathrm{ord}(U) = \mathrm{ord}(V),$$

and so for each well ordered set U, *there is exactly one ordinal number* α *such that* $U =_o \alpha$.

Conditions (12-12) and (12-13) are precisely Cantor's (12-1) and (12-3). The key, last condition (12-14) is characteristic of the von Neumann ordinal assignment, Problem **x12.4**. This is an interesting result; we formulated it as a problem only because it makes for a good one, and we will not need to appeal to it. However, it is easy to get lost in proving scores of elementary properties of ordinals, some useful, others just challenging, and the proofs from the definition are a bit confusing: it is not entirely natural to think of the membership relation as an ordering. It is good practice, at least in the beginning, to prove properties of ordinals directly from their three, basic properties isolated above, which summarize their most basic features.

12.14. Lemma (Ordinal comparison). *For any two ordinals* α, β,

$$\alpha \leq_o \beta \iff \alpha = \beta \vee \alpha \in \beta \iff \alpha \sqsubseteq \beta \iff \alpha \subseteq \beta.$$

PROOF. We give a round-robin argument of the strict versions of the claimed equivalences.

(1) $\alpha <_o \beta \Longrightarrow \alpha \in \beta$ follows immediately from (12-12) and (12-14), since the hypothesis means that $\alpha = \mathrm{ord}(U)$ and $\beta = \mathrm{ord}(V)$ with $U <_o V$.

(2) $\alpha \in \beta \Longrightarrow \alpha \sqsubsetneq \beta$ and (3) $\alpha \sqsubsetneq \beta \Longrightarrow \alpha \subsetneq \beta$ follow equally easily from (12-12) and (12-14) and we will skip them.

(4) $\alpha \subsetneq \beta \Longrightarrow \alpha <_o \beta$. The hypothesis gives us an injection from α to β (the identity!), and so $\alpha \leq_o \beta$ by Corollary **7.32**; but $\alpha =_o \beta$ implies $\alpha = \beta$ by Exercise **12.6** which contradicts the assumed, proper inclusion $\alpha \subsetneq \beta$, and so $\alpha <_o \beta$. ⊣

It is traditional to use for the ordering on ordinals the simplest notation,

$$\alpha \leq \beta \iff_{\mathrm{df}} \alpha \leq_o \beta \quad (\alpha, \beta \in \mathrm{ON}), \tag{12-15}$$

keeping in mind its equivalent characterizations in Lemma **12.14**. We summarize its properties in one, now simple result.

12.15. Theorem (The ordering of ON). (1) *The class* ON *of ordinal numbers is well ordered by the condition* $\alpha \leq \beta$, *in the following precise sense*:

$$\alpha \leq \alpha, \quad \alpha \leq \beta \,\&\, \beta \leq \gamma \Longrightarrow \alpha \leq \gamma, \quad \alpha \leq \beta \,\&\, \beta \leq \alpha \Longrightarrow \alpha = \beta,$$
$$\alpha < \beta \vee \alpha = \beta \vee \beta < \alpha,$$

and for every definite condition P,

$$(\exists \alpha \in \mathrm{ON})P(\alpha) \Longrightarrow (\exists \alpha \in \mathrm{ON})[P(\alpha) \,\&\, (\forall \beta < \alpha)\neg P(\beta)].$$

In particular, there is no infinite descending chain of ordinals,

$$\alpha_0 \geq \alpha_1 \geq \alpha_2 \geq \cdots \Longrightarrow (\exists n)[\alpha_n = \alpha_{n+1}]. \tag{12-16}$$

When $P(\alpha)$ holds for some α, we set

$$(\mu\alpha \in \mathrm{ON})P(\alpha) = \min\{\alpha \in \mathrm{ON} \mid P(\alpha)\}. \tag{12-17}$$

(2) *For each ordinal number there is a next one,*

$$S(\alpha) =_{\mathrm{df}} (\mu\beta \in \mathrm{ON})[\alpha < \beta] = \alpha \cup \{\alpha\}. \tag{12-18}$$

(3) *Each set* A *of ordinal numbers has a least upper bound,*

$$\sup A =_{\mathrm{df}} (\mu\beta \in \mathrm{ON})(\forall \alpha \in A)[\alpha \leq \beta] = \bigcup A, \tag{12-19}$$

which is the maximum of A (*if* A *has a maximum*) *and* 0 *if* $A = \emptyset$.

PROOF is left for the Problems, **x12.1** – **x12.3**. ⊣

12.16. Exercise. *For each non-empty set of ordinals* $\mathscr{E}$,

$$(\mu\alpha \in \mathrm{ON})[\alpha \in \mathscr{E}] = \bigcap \mathscr{E}.$$

The **successor ordinals** are those of the form $S(\alpha)$ and the **limit ordinals** are those which are not successors or 0, so that $\alpha < \lambda \Longrightarrow S(\alpha) < \lambda$; these are also characterized by the property

$$\text{Limit}(\lambda) \iff \lambda \neq 0 \,\&\, \lambda = \sup\{\alpha \mid \alpha < \lambda\}. \tag{12-20}$$

We can prove properties of ordinals by *transfinite induction* and define operations on them by *transfinite recursion*, as follows.

12.17. Theorem (Ordinal induction). *For every unary definite condition P,*

$$(\forall\alpha)[(\forall\xi < \alpha)P(\xi) \Longrightarrow P(\alpha)] \implies (\forall\alpha)P(\alpha).$$

PROOF. Towards a contradiction, let α be least such that $\neg P(\alpha)$; now $P(\xi)$ holds for all $\xi < \alpha$, and so $P(\alpha)$ holds by the hypothesis, which we assumed it does not. ⊣

12.18. Theorem (Ordinal recursion). *For every binary definite operation H, there exists a unary definite operation F, which satisfies the identity*

$$F(\alpha) = H(F\restriction\alpha, \alpha) \quad (\alpha \in \text{ON}). \tag{12-21}$$

Here $F\restriction\alpha$ is the function $\{(\xi, F(\xi)) \mid \xi \in \alpha\}$ obtained by restricting $F(\xi)$ to $\alpha = \{\xi \mid \xi < \alpha\}$.

Similarly, with parameters, given $H(w, \alpha, x)$, there exists $F(\alpha, x)$ such that for all α, x,

$$F(\alpha, x) = H(\{(\xi, F(\xi, x)) \mid \xi < \alpha\}, \alpha, x) \quad (\alpha \in \text{ON}).$$

PROOF of the simpler, parameter-free version.

For each β, by Corollary **11.6** on the well ordered set $(\beta, \leq_\beta)$, there exists exactly one function $f_\beta : \beta \twoheadrightarrow E_\beta$ which satisfies the identity

$$\begin{aligned} f_\beta(\alpha) &= H(f_\beta\restriction\{x \in \beta \mid x <_\beta \alpha\}, \alpha) \quad (\alpha < \beta), \\ &= H(f_\beta\restriction\alpha, \alpha), \end{aligned} \tag{12-22}$$

using the fact that $<_\beta$ coincides with $\in$ and the members of β are ordinals. We claim that

$$\text{if } \alpha < \beta \text{ and } \alpha < \gamma, \text{ then } f_\beta(\alpha) = f_\gamma(\alpha);$$

if not, then there would exist a least α for which this fails for some β and γ, and then (12-22) yields a contradiction immediately. Thus, we can set

$$F(\alpha) = f_{S(\alpha)}(\alpha),$$

so $F(\alpha) = f_\beta(\alpha)$ for any $\beta > \alpha$ and (12-22) implies the required identity for the operation F.

The proof for the version with parameters is only notationally more complex. ⊣

Using this theorem, we can define arithmetical operations on ON and study their structure. We will leave most of this for the problems, but it is worth recording here the two most basic definitions, as examples of Theorem **12.18**, and in order to have some notation available to name specific ordinals.

12.19. Theorem (Ordinal addition and multiplication). *There exist binary, definite operations $\alpha + \beta$ and $\alpha \cdot \beta$ on the ordinals which satisfy the following identities*:

$$\begin{aligned} \alpha + 0 &= \alpha, \\ \alpha + S(\beta) &= S(\alpha + \beta), \\ \alpha + \lambda &= \sup\{\alpha + \beta \mid \beta < \lambda\}, \text{ if } \mathrm{Limit}(\lambda). \end{aligned} \tag{12-23}$$

$$\begin{aligned} \alpha \cdot 0 &= 0, \\ \alpha \cdot S(\beta) &= (\alpha \cdot \beta) + \alpha, \\ \alpha \cdot \lambda &= \sup\{\alpha \cdot \beta \mid \beta < \lambda\}, \text{ if } \mathrm{Limit}(\lambda). \end{aligned} \tag{12-24}$$

PROOF. We set $\alpha + \beta = F(\beta, \alpha)$, where $F(\beta, \alpha)$ is defined by the following recursion on $\beta \in \mathrm{ON}$, with α as parameter:

$$F(\beta, \alpha) = \begin{cases} \alpha, & \text{if } \beta = 0, \\ S(F(\gamma, \alpha)), & \text{if } \beta = S(\gamma), \text{ for some } \gamma, \\ \sup\{F(\xi, \alpha) \mid \xi < \beta\}, & \text{if } \mathrm{Limit}(\beta). \end{cases}$$

We leave for Problem **x12.6** the (similar) argument for multiplication. ⊣

We have already introduced the ordinal ω in (12-7), and proved in Exercise **12.8** that $\omega = \mathrm{ord}(\mathbb{N}, \leq_{\mathbb{N}})$. The ordinals following it immediately are obviously

$$\omega + 1 = S(\omega), \ \omega + 2 = S(\omega + 1), \ \omega + 3 = S(\omega + 2), \ldots$$

and right above these comes

$$\omega + \omega = \sup\{\omega + n \mid n \in \omega\} = \omega \cdot 2. \tag{12-25}$$

This is the second limit ordinal, the first one above ω. Each $\omega \cdot n$ can be obtained by adding ω to itself n times, directly from the definition. Next comes

$$\omega^2 = \sup\{\omega \cdot n \mid n < \omega\},$$

after a while $\omega^3 = \omega^2 \cdot \omega$, etc.

Many of the properties of ordinal addition and multiplication are most easily derived from the properties of these operations on (well ordered) sets, as we defined them in **7.37** and **7.38**, using the following three exercises.

12.20. Exercise. *For every ordinal* α,

$$\alpha + 1 = \mathrm{ord}(\mathrm{Succ}(\alpha)),$$

by the definition of the successor poset Succ(P) *in* **7.16**.

12.21. Exercise. *For all ordinals* α, β,

$$\alpha + \beta = \mathrm{ord}(\alpha +_o \beta),$$

by the definition of addition of posets in **7.37**. *It follows that addition of ordinals is associative but not commutative, and that*

$$\beta < \gamma \Longrightarrow \alpha + \beta < \alpha + \gamma. \tag{12-26}$$

12.22. Exercise. *For all ordinals* α, β,

$$\alpha \cdot \beta = \mathrm{ord}(\alpha \cdot_o \beta),$$

by the definition of multiplication of posets in **7.38**. *It follows that multiplication of ordinals is associative but not commutative, and that*

$$0 < \alpha \,\&\, \beta < \gamma \Longrightarrow \alpha \cdot \beta < \alpha \cdot \gamma. \tag{12-27}$$

On the other hand, some of the properties of ordinal arithmetic are more easily shown by ordinal induction, directly from their recursive definitions:

12.23. Exercise (Distribution of $\cdot$ over $+$ to the right). *For all ordinals* α, β, γ,

$$\alpha \cdot (\beta + \gamma) = \alpha \cdot \beta + \alpha \cdot \gamma.$$

The solutions of these exercises suggest methods for establishing several additional results of ordinal arithmetic, which we will leave for the problems. Note that in addition to commutativity, many other properties of natural number arithmetic fail for ordinals, including the distribution of $\cdot$ over $+$ to the left, Problem **x12.13**.

12.24. Limits of ordinal sequences. If $\alpha_0 \leq \alpha_1 \leq \cdots$ is a non-decreasing sequence of ordinal numbers, we will write

$$\lim_n \alpha_n = \sup\{\alpha_n \mid n = 0, 1, \ldots\}.$$

The notation is useful, but we must be careful when we use it because these limits do not satisfy the usual "limit theorems" of calculus, cf. Problem **x12.9**.

Next we describe von Neumann's elegant solution of the problem of *cardinal assignment* **4.20** for well orderable sets, which is based on the fact that every one of them is equinumerous with an ordinal, Lemma **12.9**.

12.25. Definition (Von Neumann cardinals)**.** We set

$$|A| = \begin{cases} (\mu\xi \in \mathrm{ON})[A =_c \xi], & \text{if } A \text{ is well orderable}, \\ A, & \text{otherwise}, \end{cases} \tag{12-28}$$

and we assume from now on that the cardinal assignment we fixed back in **4.21** of Chapter **3** is, in fact, this one. The values of $|A|$ for well orderable A are the **von Neumann cardinals**,

$$\mathrm{Card}_v(\kappa) \iff \text{for some well orderable } A, \kappa = |A|, \tag{12-29}$$

and they are easy to characterize as the **initial ordinals**:

12.26. Exercise. $\mathrm{Card}_v(\kappa) \iff \mathrm{ON}(\kappa) \,\&\, (\forall \alpha < \kappa)[\kappa \neq_c \alpha]$, *and for every* $\kappa \in \mathrm{Card}_v$, $|\kappa| = \kappa$.

By Lemma **9.11**,

$$\mathrm{Card}_v(\kappa) \Longrightarrow \mathrm{Card}_v(\kappa^+),$$

and by Lemmas **9.18** and **9.20**, if $\mathscr{E}$ is a non-empty set of cardinals, then

$$(\forall \kappa \in \mathscr{E})\mathrm{Card}_v(\kappa) \Longrightarrow \mathrm{Card}_v(\inf_c(\mathscr{E})) \text{ and } \mathrm{Card}_v(\sup_c(\mathscr{E}));$$

in fact immediately from (12-19) and **12.16**, if $\mathscr{E}$ is a non-empty set of von Neumann cardinals, then

$$\inf_c(\mathscr{E}) = \min(\mathscr{E}) = \bigcap \mathscr{E}, \quad \sup_c(\mathscr{E}) = \sup \mathscr{E} = \bigcup \mathscr{E}.$$

The von Neumann construction provides a strong cardinal assignment on the class of well orderable sets, in the sense of **4.21**:

12.27. Exercise. *If A and B are well orderable, then*

$$A =_c |A|, \text{ and } A =_c B \iff |A| = |B|. \tag{12-30}$$

Moreover, for each set $\mathscr{E}$ of sets, the class $\{|X| \mid X \in \mathscr{E}\}$ is a set.

The most useful property of von Neumann cardinals is the simplest:

$$\text{if } \kappa, \lambda \in \mathrm{Card}_v, \text{ then } \kappa =_c \lambda \iff \kappa = \lambda;$$

this follows immediately from the last two exercises and transforms all the equinumerosities between von Neumann cardinals into equations. Finally, for well orderable cardinals, we can write

$$\kappa \cdot (\lambda + \mu) = \kappa \cdot \lambda + \kappa \cdot \mu, \quad \left(\kappa^{\lambda}\right)^{\mu} = \kappa^{\lambda \cdot \mu},$$

etc., without the annoying subscript $_c$.

12.28. Cardinals, Choice and Replacement. One can make a good case that Cantor's *units* in the intuitive description of cardinals quoted in **4.19** are modeled faithfully by the von Neumann ordinals, and the quotation

> $\overline{\overline{A}}$ grows, so to speak, out of A in such a way that from every element x of A a special unit of A arises

describes precisely the construction of $|A| = \mathrm{ord}(A) = \boldsymbol{v}[A]$ relative to some best wellordering of A, Problem **x12.28**. Whatever the value of the imagery, von Neumann's construction is certainly very useful, if only because it produces a genuine cardinal arithmetic from the calculus of equinumerosities that we have developed, at least when we assume the Axiom of Choice, so that all sets have von Neumann cardinals.

About **AC**, we have been careful to state results about cardinality without assuming it, whenever this was possible, but of course the main effect has been to make clear just how poor cardinal arithmetic is without it. The main problem is the equivalence of cardinal comparability with **AC**: we do not have much of an arithmetic if we cannot compare numbers for size, and we cannot assume comparability without (necessarily) conceding the truth of the full Axiom of Choice.

Granting **AC**, how important is the existence of "true cardinals" which satisfy (12-30) and whose construction requires not only **AC** but also the Axiom of Replacement? Not much, by any account, unless you are allergic to subscripts. Thus, it might appear that von Neumann's solution of the problem of Cardinal Assignment is primarily an exercise in mathematical elegance. There is some truth to this, but one must not draw the further conclusion that the Axiom of Replacement is unimportant for cardinal arithmetic, just because its basic identities can be established in **ZDC+AC**, as equinumerosities. The problem is that **ZDC+AC** cannot prove the existence of any cardinals above the first infinite sequence

$$\aleph_0, \aleph_1, \aleph_2, \dots ,$$

and in fact it cannot even show that the sequence $(n \mapsto \aleph_n)$ exists, Problem **xB.10**. In particular, the existence of singular cardinals cannot be shown in **ZDC+AC**, so that the whole theory of cofinality remains possibly vacuous without the Axiom of Replacement.

The upshot is that to have a decent cardinal arithmetic, you must assume both the Axioms of full Choice and Replacement, i.e., to work in a theory at least as strong as **ZFDC+AC**. It is sometimes claimed that the Principle of Foundation is also necessary for cardinal arithmetic, but this is not true—although some of the most important applications of cardinals are to the structure of von Neumann's universe $\mathcal{V}$ of pure, grounded sets.

12.29. Proposition (The alephs). *By recursion on* $\alpha \in \mathrm{ON}$, *we set*

$$\begin{aligned} \aleph_0 &= |\mathbb{N}| = \omega, \\ \aleph_{\beta+1} &= \aleph_\beta^+, \\ \aleph_\lambda &= \sup\{\aleph_\beta \mid \beta < \lambda\}, \text{ if Limit}(\lambda). \end{aligned} \tag{12-31}$$

Each $\aleph_\alpha$ *is a von Neumann cardinal,*

$$\alpha < \beta \Longrightarrow \aleph_\alpha <_c \aleph_\beta \quad (\alpha, \beta \in \mathrm{ON}),$$

and every infinite von Neumann cardinal is $\aleph_\alpha$ *for some* α.

Proof. To check first that $\alpha \mapsto \aleph_\alpha$ is strictly increasing in cardinality, fix α and (towards a contradiction) choose β least such that $\alpha < \beta$ and $\aleph_\alpha =_c \aleph_\beta$: now $\beta \neq \alpha + 1$, since $\aleph_\alpha <_c \aleph_\alpha^+ = \aleph_{\alpha+1}$; $\beta = \gamma + 1$ for some $\gamma > \alpha$ implies that $\aleph_\alpha <_c \aleph_\gamma <_c \aleph_\beta$ (by the choice of β), which is a contradiction; and if β is limit, then $\alpha+1 < \beta$, and so $\aleph_\alpha <_c \aleph_{\alpha+1} <_c \aleph_\beta$, which is also a contradiction.

It follows that each $\aleph_\alpha$ is a von Neumann cardinal: this is immediate for 0 and successor ordinals, and for limit λ, if $\aleph_\lambda$ is not a cardinal, then $\aleph_\lambda \leq_c \aleph_\beta$ for some $\beta < \lambda$, which contradicts $\aleph_\beta <_c \aleph_\lambda$.

Finally, to show that every von Neumann cardinal is an aleph, let (towards a contradiction) κ be the least counterexample. Directly from the definition, $\kappa \neq \omega$ and $\kappa \neq_c \lambda^+$ for any ordinal $\lambda < \kappa$, since that implies $\kappa = |\lambda|^+$, $|\lambda|$ is an aleph by the choice of κ, and then κ is also an aleph. Let

$$\beta = \{\alpha \leq \kappa \mid \aleph_\alpha <_c \kappa\}.$$

This is an ordinal, since it is closed under $<$; it is a limit ordinal, because $\aleph_\alpha < \kappa \Longrightarrow \aleph_{\alpha+1} < \kappa$; and by its definition,

$$\alpha < \beta \Longrightarrow \aleph_\beta < \kappa.$$

On the other hand, if λ is a cardinal and $\lambda < \kappa$, then $\lambda = \aleph_\alpha$ for some $\alpha < \beta$, by the choice of κ, so that finally

$$\kappa = \sup\{\aleph_\alpha \mid \alpha < \beta\} = \aleph_\beta. \qquad \dashv$$

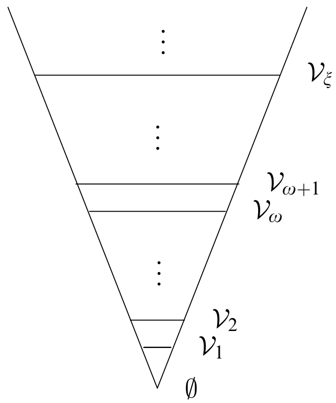

FIGURE **12.3**. Logarithmic rendition of the pure, grounded sets.

The operation $\alpha \mapsto \aleph_\alpha$ supplies a useful notation for cardinal arithmetic, especially when we accept the Axiom of Choice.

12.30. Exercise. *The Axiom of Choice* **AC** *is equivalent to the proposition that "every infinite cardinal is an aleph",*

$$\mathbf{AC} \iff (\forall \text{ infinite } A)(\exists \alpha \in \mathrm{ON})[A =_c |A| = \aleph_\alpha].$$

12.31. Exercise. (**AC**) *The Generalized Continuum Hypothesis is equivalent to the cardinal identity*

$$\mathbf{GCH} \iff 2^{\aleph_\alpha} = \aleph_{\alpha+1} \quad (\alpha \in \mathrm{ON}).$$

12.32. The Cumulative Hierarchy of Pure, Grounded Sets. For each ordinal α we define the set $\mathcal{V}_\alpha$ by the following recursion on ON:

$$\begin{aligned} \mathcal{V}_0 &= \emptyset, \\ \mathcal{V}_{\alpha+1} &= \mathcal{P}(\mathcal{V}_\alpha), \\ \mathcal{V}_\lambda &= \bigcup\nolimits_{\alpha<\lambda} \mathcal{V}_\alpha, \quad \text{if Limit}(\lambda). \end{aligned}$$

The **von Neumann universe** is the union of all the $\mathcal{V}_\alpha$'s

$$\mathcal{V} =_{\mathrm{df}} \bigcup\nolimits_{\alpha\in\mathrm{ON}} \mathcal{V}_\alpha = \{x \mid \text{for some } \alpha \in \mathrm{ON}, x \in \mathcal{V}_\alpha\}, \tag{12-32}$$

and on it we define the **rank** operation by

$$\mathrm{Rank}(x) = (\mu\alpha \in \mathrm{ON})[x \in \mathcal{V}_{\alpha+1}] \quad (x \in \mathcal{V}). \tag{12-33}$$

We have already used the symbol $\mathcal{V}$ to denote the class of pure grounded sets, because of the next result.

12.33. Theorem. (1) *Each* $\mathcal{V}_\alpha$ *is a pure, transitive, grounded set, and*

$$\alpha \le \beta \Longrightarrow \mathcal{V}_\alpha \subseteq \mathcal{V}_\beta.$$

(2) *If* λ *is a limit ordinal,* $\lambda \ge \omega \cdot 2$*, then* $\mathcal{V}_\lambda$ *is a Zermelo universe.*

(3) *For each pure set A, $A \subseteq \mathcal{V} \Longrightarrow A \in \mathcal{V}$.*
(4) *The von Neumann universe $\mathcal{V}$ comprises the pure, grounded sets.*

PROOF. The arguments for Parts (1) and (2) are those we used to prove the corresponding properties of the basic closure sets $M(I)$ with transitive I in Lemma **11.15**, and we will not repeat them.

(3) Assume that $A \subseteq \mathcal{V}$ and let $\text{Rank}[A] = \{\text{Rank}(x) \mid x \in A\}$ be the image of the rank operation on A. This is a set of ordinals, so there exists some ordinal κ strictly above its members,

$$x \in A \Longrightarrow \text{Rank}(x) < \kappa \Longrightarrow x \in \mathcal{V}_\kappa,$$

and (using the purity of A),

$$A = \{x \in \mathcal{V}_\kappa \mid x \in A\} \in \mathcal{V}_{\kappa+1}.$$

(4) We use the contrapositive of (3), which says that for pure A,

$$A \notin \mathcal{V} \Longrightarrow (\exists x \in A)[x \notin \mathcal{V}]. \tag{12-34}$$

Suppose first that M is pure, transitive and grounded but $M \notin \mathcal{V}$. The transitivity of M and (12-34) yield

$$(\forall x \in M \setminus \mathcal{V})(\exists y \in M \setminus \mathcal{V})[y \in x],$$

and then **DC** gives us a descending $\in$-chain which proves M ill founded, contradicting the hypothesis. Thus, every pure, transitive, grounded set is in $\mathcal{V}$, so for every pure, grounded set A, $\text{TC}(A) \in \mathcal{V}$, and then $A \in \mathcal{V}$, since $A \in \text{TC}(A)$ and $\mathcal{V}$ is transitive. $\dashv$

Recall that from Theorem **11.34**, $\mathcal{V}$ is a **ZFDC**-universe, in fact a **ZFC**-universe if we assume **AC**.

12.34. The naive notion of pure, grounded set. In discussing the Principle of Foundation in **11.32**, we argued that the definition of the least Zermelo universe $\mathcal{Z}$ and the proofs of its basic properties can be understood directly and naively, as we usually understand mathematics, and that they carry considerable force of persuasion as an intuitive conception of "set" which justifies the axioms of **ZDC+AC** and the Principle of Foundation. In the same vein, we can argue that the "construction" of von Neumann's universe $\mathcal{V}$ in **12.32** and Theorem **12.33** can be understood directly and naively outside the details of any specific axiomatization, and that it puts forward a natural, intuitive conception of "pure, grounded set" which justifies the axioms of **ZFC**. It is worth looking into these arguments a little closer.

The construction of $\mathcal{Z}$ begins with the infinite set

$$\mathbb{N}_0 = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \dots\}$$

and iterates the powerset operation $\mathcal{P}(A)$ infinitely many times. Since the "infinity" of iterations involved is no more and no less than that embodied by $\mathbb{N}_0$, we can say that *to understand $\mathcal{Z}$ we must understand two infinitary things: the set* $\mathbb{N}_0$ (basically the natural numbers) *and the powerset operation.*

The construction of $\mathcal{V}$ starts (literally) with nothing, the empty set, but it proceeds to iterate the powerset operation $\mathcal{P}(A)$ through all the ordinals. On the same analysis, it is fair to say that *to understand $\mathcal{V}$ we must understand the class of ordinals* ON *and the powerset operation*. One may attempt to speak eloquently about the ordinals and justify them, as one might try to justify the natural numbers or the powerset operation. It should be clear, however, that the ordinals represent a separate and different new ingredient in our intuitive understanding of $\mathcal{V}$, they cannot be reduced to $\mathbb{N}_0$ and the taking of powersets. From this point of view, the justification of the axioms of **ZFC** which we find in this intuitive construction is considerably weaker than the justification of **ZDC**+**AC** we get from contemplating $\mathcal{Z}$.

In Appendix **B** we will consider alternative set universes, including some which contain both atoms and ill founded sets, and in more advanced textbooks one can find a multitude of fascinating models of set theory constructed (primarily) by extensions and combinations of Gödel's constructibility and Cohen's forcing. Part of the reason we have worked here in the weak systems of **ZDC** and **ZFDC** is to ensure that the elementary results of axiomatic set theory we have established apply directly to (essentially) all these models. These models, however, are all constructed starting with some given model of **ZFC**, and it is not clear how to produce for any of them independent, intuitive notions of **what sets are**, which justify directly the axioms they satisfy without also justifying the axioms of **ZFC**.

It appears that (as of now), the intuitive conception of **pure, grounded set**, which is gleaned from an informal analysis of **12.32** and **12.33**, is by far the best replacement we have for Cantor's unfettered (and self-contradictory) notion of "collection into a whole of definite and separate objects".

12.35. About atoms and applications. In **3.25** we argued (with Zermelo) that it is useful to allow atoms in axiomatic set theory, so that our theorems apply directly to sets of planets or frogs, as well as to the pure, grounded sets made up out of nothing. Can the universally accepted, standard theory **ZFC** which does not allow atoms justify the applications of set theory? There are two good responses to this.

First, we can *model* physical objects and relations among them by *structures* made up of pure sets, much as we model (up to isomorphism) ordered pairs, functions, the natural numbers, etc. For example, to study the behavior of a system of heavenly bodies $P_1, \ldots, P_k$ interacting with each other under gravity, we might represent each of them by a function $\overline{P}_i : \mathbb{R} \to \mathbb{R}^7$, which assigns to each real number $t \in \mathbb{R}$ the mass, position and velocity of P_i, relative to some fixed coordinate system and units. The laws of gravity and motion will then determine these functions; and the physicist does not care whether the set theoretic objects which model the functions $\overline{P}_1, \ldots, \overline{P}_k$ are pure or not—what matters are the relations among these functions, which can then be interpreted as relations among the planets and checked against reality by observation.

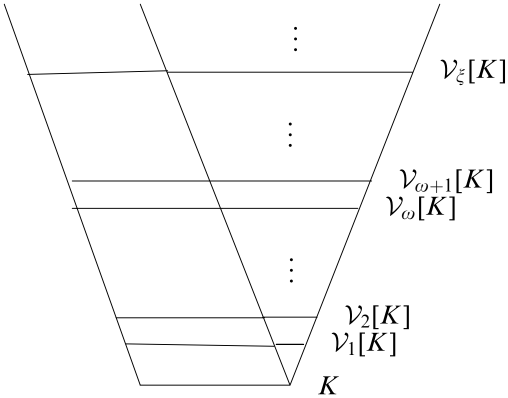

FIGURE **12.4**. Rendition of the grounded sets over K.

Second, if we value the ability to talk directly about planets within set theory, we can allow a class of atoms K and replace $\mathcal{V}$ by the class $\mathcal{V}[K]$ of grounded sets supported by K defined in **11.38**. This is a **ZFDC**-universe by Problem **x11.22**; it satisfies the Axiom of Choice;, and, in the interesting case, when K is a set, looks very much like and has essentially all the properties of $\mathcal{V}$, and by the same proofs, cf. Problem **x12.32**. It can be constructed in **ZFDC** by the ordinal recursion

$$\begin{aligned}\mathcal{V}_0[K] &= K,\\ \mathcal{V}_{\alpha+1}[K] &= \mathcal{P}(\mathcal{V}_\alpha[K]),\\ \mathcal{V}_\lambda[K] &= \textstyle\bigcup_{\alpha<\lambda}\mathcal{V}_\alpha[K], \quad \text{if Limit}(\lambda),\\ \mathcal{V}[K] &= \textstyle\bigcup_\alpha \mathcal{V}_\alpha[K].\end{aligned}$$

The main point is that the physicist does not care about the difference between these two approaches, and probably cannot even see it: because all that matters for the application of mathematics (in the example) are the functions $\overline{P}_i$ which codify the properties of the planets, just as all that matters about the natural numbers is that they form a Peano system—what they actually *are* is of no consequence. And so, ultimately, it is most useful to "ban" atoms and accept the simpler **ZFC** as the "standard" axiomatic set theory, which is what is done without exception in all advanced work on our topic.

Problems for Chapter 12

x12.1. Prove (1) of Theorem **12.15**.

x12.2. Prove (2) of Theorem **12.15**.

x12.3. Prove (3) of Theorem **12.15**.

***x12.4. Characterization of von Neumann ordinals**. Suppose $\phi(V)$ is a definite operation which assigns well ordered sets to well ordered sets and which satisfies the following three conditions:

$$V =_o \phi(V),$$
$$U \leq_o V \Longrightarrow \phi(U) \sqsubseteq \phi(V),$$
$$\text{Field}(\phi(V)) = \{\text{Field}(\phi(U)) \mid U <_o V\}.$$

Prove that $\text{ord}(V) = \alpha \Longrightarrow \phi(V) = (\alpha, \leq_\alpha)$.

x12.5. The class ON is not a set.

x12.6. Justify the definition of ordinal multiplication in Theorem **12.19**.

x12.7. For all ordinals $\alpha, \beta, \gamma, \delta$:

$$0 + \alpha = \alpha, \text{ and } \omega \leq \alpha \Longrightarrow 1 + \alpha = \alpha,$$
$$0 < \beta \Longrightarrow \alpha < \alpha + \beta,$$
$$\alpha \leq \beta \,\&\, \gamma \leq \delta \Longrightarrow \alpha + \gamma \leq \beta + \delta,$$
$$\alpha \leq \beta \,\&\, \gamma < \delta \Longrightarrow \alpha + \gamma < \beta + \delta.$$

Show also that, in general,

$$\alpha < \beta \text{ does not imply } \alpha + \gamma < \beta + \gamma.$$

x12.8. If $\alpha_0 < \alpha_1 < \cdots$ is a strictly increasing sequence of ordinals, then the limit $\lim_n \alpha_n$ is a limit ordinal.

x12.9. Give examples of strictly increasing sequences of ordinals such that

$$\lim_n(\alpha_n + \beta) \neq \lim_n \alpha_n + \beta,$$
$$\lim_n(\alpha_n + \beta_n) \neq \lim_n \alpha_n + \lim_n \beta_n.$$

x12.10. For all ordinals $\alpha, \beta, \gamma, \delta$:

$$0 \cdot \alpha = 0$$
$$0 < \alpha \,\&\, 1 < \beta \Longrightarrow \alpha < \alpha \cdot \beta$$
$$\alpha \leq \beta \,\&\, \gamma \leq \delta \Longrightarrow \alpha \cdot \gamma \leq \beta \cdot \delta$$
$$0 < \alpha \leq \beta \,\&\, \gamma < \delta \Longrightarrow \alpha \cdot \gamma < \beta \cdot \delta.$$

Show also that even when $\gamma > 0$, in general,

$$\alpha < \beta \text{ does not imply } \alpha \cdot \gamma < \beta \cdot \gamma.$$

x12.11 (Cancellation laws)**.** For all ordinals α, β, γ,

$$\alpha + \beta < \alpha + \gamma \implies \beta < \gamma,$$
$$\alpha + \beta = \alpha + \gamma \implies \beta = \gamma,$$
$$\alpha \cdot \beta < \alpha \cdot \gamma \implies \beta < \gamma,$$
$$0 < \alpha \,\&\, \alpha \cdot \beta = \alpha \cdot \gamma \implies \beta = \gamma.$$

Show also that, in general,

$$0 < \alpha \,\&\, \beta \cdot \alpha = \gamma \cdot \alpha \text{ does not imply } \beta = \gamma.$$

x12.12. For all $\alpha \geq \omega$ and $n < \omega$:

$$\begin{aligned} n + \alpha &= \alpha, \\ (\alpha + 1) \cdot n &= \alpha \cdot n + 1 \quad (n > 1), \\ (\alpha + 1) \cdot \omega &= \alpha \cdot \omega. \end{aligned}$$

x12.13. Give an example of three ordinals α, β, γ for which

$$(\alpha + \beta) \cdot \gamma \neq \alpha \cdot \gamma + \beta \cdot \gamma.$$

HINT. Use the preceding problem.

x12.14. If $n < m$, then $\omega^n + \omega^m = \omega^m$.

x12.15 (Ordinal subtraction)**.** If $\alpha \leq \gamma$, then there exists exactly one β such that $\gamma = \alpha + \beta$.

x12.16. Every ordinal $\alpha < \omega^2$ can be written uniquely in the form

$$\alpha = \omega \cdot x + y \qquad (x, y < \omega).$$

***x12.17.** For any ordinal $\alpha < \omega^N$, there are unique $n < N, x < \omega, \beta < \omega^n$ such that $\alpha = \omega^n \cdot x + \beta$. HINT. Choose n largest such that $\omega^n \leq \alpha$, and x largest such that $\omega^n \cdot x \leq \alpha$.

x12.18. For any $N > 0$, every ordinal $\alpha < \omega^N$ can be written uniquely in the form

$$\alpha = \omega^{n_1} \cdot x_1 + \omega^{n_2} \cdot x_2 + \cdots + \omega^{n_s} \cdot x_s + x_{s+1}, \tag{12-35}$$

where $N > n_1 > n_2 > \cdots > n_s$ and $x_1, \ldots, x_{s+1} < \omega$. HINT. Use induction on N and the preceding problem.

12.36. Definition (Normal operations)**.** A unary, definite operation F on the ordinals is **normal** if it is strictly increasing

$$\alpha < \beta \Longrightarrow F(\alpha) < F(\beta),$$

and continuous at limit ordinals, i.e.,

$$F(\lambda) = \sup\{F(\beta) \mid \beta < \lambda\}, \text{ if Limit}(\lambda).$$

x12.19. If $F : \mathrm{ON} \to \mathrm{ON}$ is a normal operation, then for every α, $\alpha \leq F(\alpha)$, and for every limit ordinal λ, $F(\lambda)$ is a limit ordinal.

x12.20. For any fixed α, the operations

$$S_\alpha(\beta) = \alpha + \beta, \quad P_\alpha(\beta) = \alpha \cdot \beta$$

of "adding to" and "multiplying on the left by" α are normal.

x12.21. The composition $F(\alpha) = G(H(\alpha))$ of normal operation is normal.

x12.22. Define *ordinal exponentiation* α^β (for $\alpha > 1$), so that

$$\alpha^0 = 1,$$
$$\alpha^{S(\beta)} = \alpha^\beta \cdot \alpha,$$
$$\alpha^\lambda = \sup\{\alpha^\beta \mid \beta < \lambda\}, \text{ if Limit}(\lambda),$$

and prove the following:

(1) If $\beta < \gamma$, then $\alpha^\beta < \alpha^\gamma$.

(2) For each $\alpha > 1$, the operation $E_\alpha(\beta) = \alpha^\beta$ is normal.

(3) $\alpha^{(\beta+\gamma)} = \alpha^\beta \cdot \alpha^\gamma$.

(4) $\left(\alpha^\beta\right)^\gamma = \alpha^{\beta\cdot\gamma}$.

HINT. Exercises **x12.20** and **x12.21** and (2) simplify considerably the proofs of (3) and (4).

***x12.23.** If $\alpha > 0$, then there is a largest β such that $\omega^\beta \leq \alpha$, and for that β, $\alpha = \omega^\beta + \gamma$ with some $\gamma < \alpha$. HINT. Use Problem **x12.15** to get the γ, and prove that $\gamma < \omega^{\beta+1}$, which contradicts $\gamma \geq \alpha$.

***x12.24.** (1) If $\beta < \gamma$, then $\omega^\beta + \omega^\gamma = \omega^\gamma$.

(2) Every ordinal $\alpha > 0$ can be written uniquely in the form $\alpha = \omega^\beta + \gamma$ with $\gamma < \alpha$.

***x12.25** (Cantor normal form)**.** Every ordinal $\alpha > 0$ can be written uniquely in the form of a finite sum of non-increasing powers of ω,

$$\alpha = \omega^{\beta_1} + \omega^{\beta_2} + \cdots + \omega^{\beta_s} \quad (\beta_1 \geq \beta_2 \geq \cdots \geq \beta_s), \tag{12-36}$$

or, equivalently,

$$\alpha = \omega^{\beta_1} \cdot n_1 + \omega^{\beta_2} \cdots n_2 + \cdot + \omega^{\beta_t} \cdot n_t$$
$$(\beta_1 > \beta_2 > \cdots > \beta_t, n_i < \omega, n_i \neq 0). \tag{12-37}$$

HINT. For the uniqueness, prove first by induction on s that

if $\beta \geq \beta_1 \geq \beta_2 \geq \cdots \geq \beta_s$ and $\gamma = \omega^{\beta_1} + \omega^{\beta_2} + \cdots + \omega^{\beta_s}$,

then $\gamma < \omega^\beta + \gamma$.

If ε_0 is the least fixed point of the normal operation $(\alpha \mapsto \omega^\alpha)$, then its Cantor canonical form is the useless

$$\varepsilon_0 = \omega^{\varepsilon_0} = \omega^{\omega^{\varepsilon_0}} = \cdots$$

But the ordinals less than ε_0 have non-trivial Cantor canonical forms which provide simple and (sometimes) useful representations of them.

x12.26. Find the Cantor canonical form of $\omega \cdot (\omega^\omega + 1) + (\omega^\omega + 1) \cdot \omega$.

***x12.27.** The only ordinals which belong to the least Zermelo universe $\mathcal{Z}$ are the finite ones.

x12.28. If $\leq$ is a best wellordering of A, then $|A| = \boldsymbol{\nu}_U[A]$, i.e., $|A|$ is the ordinal assigned to the well ordered set $(A, \leq)$ by its von Neumann map.

x12.29. The class Card_v of von Neumann cardinal numbers is not a set.

x12.30. (**AC**) The definite operation $\beth_\alpha$ (read "Beth-alpha") is defined by the following recursion on ON:

$$\begin{aligned} \beth_0 &= \aleph_0 = |\mathbb{N}| = \omega, \\ \beth_{\beta+1} &= 2^{\beth_\beta}, \\ \beth_\lambda &= \sup\{\beth_\beta \mid \beta < \lambda\}, \text{ if Limit}(\lambda). \end{aligned} \tag{12-38}$$

Prove that for every ordinal α,

$$|\mathcal{V}_{\omega+\alpha}| = \beth_\alpha.$$

x12.31. For every ordinal α, $\mathrm{Rank}(\alpha) = \alpha$.

x12.32. For each set of atoms K, the rank hierarchy $(\alpha \mapsto \mathcal{V}_\alpha[K])$ has the following properties.

(1) Each $\mathcal{V}_\alpha[K]$ is a transitive, grounded set, supported by K, and

$$\alpha \leq \beta \Longrightarrow \mathcal{V}_\alpha[K] \subseteq \mathcal{V}_\beta[K].$$

(2) If λ is a limit ordinal, $\lambda \geq \omega \cdot 2$, then $\mathcal{V}_\lambda[K]$ is a Zermelo universe.

(3) For each set A supported by K, $A \subseteq \mathcal{V}[K] \Longrightarrow A \in \mathcal{V}[K]$.

(4) The universe $\mathcal{V}[K]$ comprises the grounded sets supported by K.

x12.33. For each set of atoms K and each ordinal α,

$$\mathcal{V}_\alpha = \mathcal{V}_\alpha[K] \cap \{x \mid x \text{ is pure}\}.$$

***x12.34.** Suppose $\pi : K \rightarrowtail\!\!\!\!\twoheadrightarrow K$ is a permutation of a set of atoms K. Prove that there is a unique extension $\pi^* : \mathcal{V}[K] \rightarrowtail\!\!\!\!\twoheadrightarrow \mathcal{V}[K]$ of π, which is an **automorphism** of $\mathcal{V}[K]$, i.e.,

$$x \in y \iff \pi^*(x) \in \pi^*(y) \qquad (x, y \in \mathcal{V}[K]).$$

12.37. Definition (Closed unbounded classes)**.** A class of ordinals $M \subseteq \mathrm{ON}$ is **unbounded** if

$$(\forall \xi \in \mathrm{ON})(\exists \alpha \in \mathrm{ON})[\xi < \alpha \,\&\, \alpha \in M];$$

and it is **closed** if for every set of ordinals $A \neq \emptyset$,

$$A \subseteq M \Longrightarrow \sup A \in M.$$

12.38. Exercise. *If M is a closed class of ordinals and*

$$\alpha_0 < \alpha_1 < \cdots \qquad (\alpha_n \in M)$$

is an increasing sequence of ordinals in M, then $\lim_n \alpha_n \in M$.

***x12.35.** If M_1 and M_2 are closed, unbounded classes of ordinals, then their intersection $M_1 \cap M_2$ is closed and unbounded.

*x12.36.** Every normal operation on the ordinals has a fixed point, i.e., for some ordinal, α, $F(\alpha) = \alpha$; in fact, the class

$$\text{fp}(M) = \{\alpha \in \text{ON} \mid F(\alpha) = \alpha\}$$

of fixed points of F is closed and unbounded. HINT. To get one fixed point, let $\alpha_0 = 0$, $\alpha_{n+1} = F(\alpha_n)$, and show that $F(\lim_n \alpha_n) = \lim_n \alpha_n$.

x12.37. (1) There exists a von Neumann cardinal κ, such that

$$\kappa = \aleph_\kappa.$$

(2) (**AC**) There exists a von Neumann cardinal λ, such that

$$\lambda = \beth_\lambda.$$

12.39. Definition. Suppose $\alpha \leq \beta$ are infinite limit ordinals. A function $f : \alpha \to \beta$ is **cofinal** if

$$\sup\{f(\xi) \mid \xi < \alpha\} = \beta.$$

The identity $(\xi \mapsto \xi)$ is a cofinal function on every limit ordinal, for example, but $(n \mapsto \aleph_n)$ is also cofinal, from ω to $\aleph_\omega$.

The next problem establishes a useful characterization of the cofinality operation, defined in **9.23**, which is often taken as its definition.

x12.38. For each von Neumann cardinal κ,

$$\text{cf}(\kappa) = \min\{\alpha \mid \text{there exists a cofinal } f : \alpha \to \kappa\}.$$

x12.39. Prove that for all von Neumann cardinals $\lambda \leq \kappa$, there exists a cofinal function $f : \lambda \to \kappa$ if and only if $\text{cf}(\lambda) = \text{cf}(\kappa)$.

x12.40. For every regular λ, $\text{cf}(\aleph_\lambda) = \lambda$, so there exists cardinals of every regular cofinality.

*x12.41.** There exist singular cardinals of every regular cofinality.

x12.42. For every von Neumann cardinal λ with $\text{cf}(\lambda) \geq \aleph_1$, the set $\mathcal{V}_\lambda$ is a Zermelo universe which further satisfies the following special case of the Replacement Axiom: if F is a definite operation and $x \in \mathcal{V}_\lambda \Longrightarrow F(x) \in \mathcal{V}_\lambda$, then the image $F[A]$ of every countable $A \in \mathcal{V}_\lambda$ is also in $\mathcal{V}_\lambda$.

12.40. Definition. (**AC**) An uncountable cardinal number κ is **strongly inaccessible** if it is regular and

$$\lambda < \kappa \Longrightarrow 2^\lambda < \kappa.$$

*x12.43.** (**AC**) If κ is strongly inaccessible, then $\mathcal{V}_\kappa$ is a **ZFC**-universe.

*x12.44.** (**AC**) If a pure and grounded set M is a **ZFDC**-universe, then $M = \mathcal{V}_\kappa$, for a strongly inaccessible cardinal κ.

12.41. Frege cardinals. We have followed Cantor in his approach to the theory of cardinal numbers, by which the property

$$A =_c |A| \tag{12-39}$$

is most fundamental. There is another approach due to Frege, which takes $|A|$ to be not a set of "units" equinumerous with A, but the abstract notion of "being equinumerous with A". Frege understands "1", for example, as the common property of all singletons. To model this idea in set theory, it is not important to define $|A|$ so that it is equinumerous with A, in fact, it is not even necessary for $|A|$ to be a set! The only important property of cardinals is the last one,

$$A =_c B \iff |A| = |B|, \tag{12-40}$$

which (in effect) makes the operation $|A|$ a "determining surjection" of the "equivalence condition" $=_c$, with the cardinal numbers as the "quotient class", in the natural extension to classes of the terminology in **x4.5**. Frege tried to capture this idea by setting

$$|A| = \{X \mid X =_c A\}, \tag{12-41}$$

but the class $\{X \mid X =_c A\}$ is not a set (when $A \neq \emptyset$, easily) and the (necessary for the theory) assumption that it is led Frege to a contradiction.

Von Neumann cardinals reconcile the Cantor and Frege approaches by satisfying both (12-39) and (12-40), but their definition depends on both the Axioms of Choice and Replacement. Problem **x12.46** describes another approach, due to Scott, which succeeds in defining *Frege cardinals* without the Axiom of Choice, but (essentially) only for pure, grounded sets. Scott's construction is important, not so much for rescuing Frege cardinals (since little cardinal arithmetic can be done without **AC** anyway), but for the simplicity and elegance of the method, which has many uses beyond the present one. First we describe Scott's general method, and then its application to Frege cardinals.

12.42. Definition. An **equivalence condition** on a class A is any binary, definite condition $\sim$ which has the properties of an equivalence relation, i.e., for all objects $x, y, z \in A$

$$x \sim x,\ x \sim y \Longrightarrow y \sim x,\ x \sim y \,\&\, y \sim z \Longrightarrow x \sim z.$$

A unary, definite operation F is **determining for** $\sim$ if

$$x \sim y \iff F(x) = F(y) \quad (x, y \in A);$$

the class of values of F for arguments in A is the **quotient** (class) of A by $\sim$ determined by F,

$$F[A] =_{\text{df}} \{F(x) \mid x \in A\}.$$

For example, the condition $=_o$ of similarity is an equivalence condition on the class of well ordered sets, and the von Neumann ordinal assignment $\text{ord}(U)$ is a determining operation for it, with quotient the class ON of ordinals. The condition $=_c$ of equinumerosity is an equivalence condition on the class of well orderable sets, and the von Neumann cardinal $|A|$ operation is determining for it, with quotient the class of von Neumann cardinals.

We can think of an equivalence condition $\sim$ on a class A very much as if A were a set and $\sim \subseteq A \times A$ an ordinary equivalence relation on it. There is no easy way to define a determining operation for $\sim$, however, because the classical construction of equivalence classes **4.12** truly leads to "classes" which need not be sets in this case: this is the problem with Frege's definition of the number 1 above.

x12.45. (Scott) Suppose $\sim$ is an equivalence condition on a class A of pure, grounded sets, and for each $x \in A$ let

$$\begin{aligned}\rho(x) &=_{\mathrm{df}} (\mu\alpha \in \mathrm{ON})(\exists y \in \mathcal{V}_\alpha)[y \sim x],\\ F(x) &=_{\mathrm{df}} \{y \in \mathcal{V}_{\rho(x)} \mid y \sim x\}.\end{aligned}$$

Prove that F is a determining operation for $\sim$ on A.

x12.46. (Scott) Define the **Scott cardinal** $|A|_s$ of every set A which is equinumerous with a pure, grounded set, so that for all such sets A and B,

$$A =_c B \iff |A|_s = |B|_s.$$